MAST20026 REAL ANALYSIS

Sequences: Monotone and Cauchy

6A

A Note on the Solutions: This solution sheet is intended for you verify correctness of your work during the tutorial. The solutions here do not necessarily meet the standards for exposition expected on assignment submissions.

1. **The Space of All Sequences**

   1: $(f_n) = ((-1)^n n)$

   2: $(f_n) = ((-1)^n)$

   3: $(f_n) = \left(\frac{(-1)^n}{n}\right)$

   4: $(f_n) = \left(\frac{1}{n}\right)$

   5: $(f_n) = (n)$

   Why is there no region for bounded monotone sequences that are not Cauchy?

   Answer: Every bounded monotone sequences converges. All convergent sequences are Cauchy. Therefore every bounded monotone sequence is Cauchy.

   Why is there no region for Cauchy sequences that are not bounded?

   Answer: Every Cauchy sequence is bounded. You can prove this in very similar fashion to how you prove that every convergent sequence is bounded.

   In which region(s) would you find convergent sequences?

   Answer: regions 3+4

2. **Verifying a Sequence is not Cauchy: the long way**

   In this exercise we will use the definition of Cauchy to show that the sequence given by the formula $f_n = \sqrt{n}$ is not Cauchy.

   (a) For every $\epsilon > 0$, there exists $M \in \mathbb{N}^+$ so that $|\sqrt{j} - \sqrt{k}| < \epsilon$ whenever $j, k > M$.

   (b) We want to know for which $j > M$ we have $\left|\sqrt{j} - \sqrt{M+1}\right| < 1$. Since $j > M$, we can drop the absolute value signs and rearrange the inequality to conclude that

   $$\left|\sqrt{j} - \sqrt{M+1}\right| < 1 \quad \Longrightarrow \quad j < \left(1 + \sqrt{M+1}\right)^2$$

   (c) From Part (b), we see that this is false, for example, when $\epsilon = 1$. No matter what we choose for M, there is $j > M$ such that $\left|\sqrt{j} - \sqrt{M+1}\right| > 1$.

   (d) The sequence given by $f_n = \sqrt{n}$ is not bounded. Since every Cauchy sequence is bounded, it cannot be Cauchy.

3. **Recursive Sequences**

   All of the sequences we have studied so far have given a formula for f_n as an explicit function of n. However, one can define a sequence *recursively*[1]. Let (a_n) be the sequence so that $a_1 = 0$ and $a_n = \sqrt{2 + a_{n-1}}$ for all $n \geq 2$.

   (a) The first few terms are

   $$\left(0, 2, \sqrt{2+\sqrt{2}}, \sqrt{2+\sqrt{2+\sqrt{2}}}, \sqrt{2+\sqrt{2+\sqrt{2+\sqrt{2}}}}, \ldots\right)$$

   This sequence seems to be increasing. (Technically, this needs a proof by induction, see Part (d).)

   (b) For $n = 1$ we have $a_1 = 0 < 2$. And so $0 \leq a_1 < 2$.

   Let $k \geq 1$ and assume $0 \leq a_k < 2$. We have

   $$a_{k+1} = \sqrt{2 + a_k} < \sqrt{2+2} = 2.$$

   Further, by the definition of square roots, $\sqrt{2 + a_k} \geq 0$. Therefore $0 \leq a_{k+1} < 2$.

   The result now follows by induction.

[1] See footnote 1 for the definition of recursion.

(c) We must show a_n is monotone.

(d) For $n = 2$, we have $a_{n-1} = 0$ and $a_n = \sqrt{2}$, therefore $a_n > a_{n-1}$.

Let $k \geq 2$ and assume $a_k > a_{k-1}$.

We have $a_{k+1} = \sqrt{2 + a_k}$ and $a_k = \sqrt{2 + a_{k-1}}$. Since $a_k > a_{k-1}$, we have $\sqrt{2 + a_k} > \sqrt{2 + a_{k-1}}$. Therefore, $a_{k+1} > a_k$.

The result follows by induction.

Since (a_n) is bounded and monotone, it follows that (a_n) converges.

(Notice that you have managed to show something converges without having to make an $\epsilon - M$ argument!)

4. **A contractive sequence**

Consider the sequence (f_n) defined recursively by $f_1 = 1$, $f_2 = 2$, and $f_n = \frac{1}{2}(f_{n-2} + f_{n-1})$ for all $n \geq 3$.

(a) For $n = 1$ we have $f_1 = 1$. And so $1 \leq f_1 \leq 2$.

Let $k \geq 1$ and assume $1 \leq f_n \leq 2$ for all $n < k$. We have

$$f_{k+1} = \frac{1}{2}(f_{k-1} + f_k) \leq \frac{1}{2}(2 + 2) = 2.$$

Therefore $1 \leq f_{k+1} \leq 2$.

The result follows by induction.

(b) No. We know that $f_1 = 1$ and $f_2 = 2$ and we compute $f_3 = 3/2$. Thus $f_1 < f_2$ but $f_2 > f_3$. It follows that the sequence is neither monotone increasing nor monotone decreasing.

(Note that this behavior persists. It is not hard to prove by induction that or each integer $k \geq 1$, $f_{2k-1} < f_{2k}$ but $f_{2k} > f_{2k+1}$.)

(c) We compute explicitly

$$|f_{n+2} - f_{n+1}| = \left|\frac{1}{2}(f_n + f_{n+1}) - f_{n+1}\right| = \left|\frac{1}{2}(f_{n+1} - f_n)\right| = \frac{1}{2}|f_{n+1} - f_n|$$

(d) We have shown that this sequence is *contractive* (with $c = 1/2$). Therefore, the sequence converges. We can try to use the Algebra of Limits Theorem to fine the limit L, but all it will tell us is that $L = \frac{1}{2}(L + L)$, which is true but useless. In fact, the limit is $L = 5/3$.

5. $\mathbf{1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 2}$

Let $f_n = 1 + \displaystyle\sum_{k=1}^{n} \frac{1}{2^k}$.

(a) $(3/2, 7/4, 15/8, \ldots)$

(b) Recall that for a geometric series with ratio r and starting term a we have

$$a + ar + ar^2 + \cdots + ar^k = a\frac{1 - r^{k+1}}{1 - r}$$

And so

$$f_n = \frac{1 - \frac{1}{2^{n+1}}}{\frac{1}{2}} = 2 - \frac{1}{2^n}$$

Therefore, $0 < f_n < f_{n+1} < 2$ for all $n \in \mathbb{N}^+$.

(c) From Part (b), we have that 2 is an upper bound of $\{f_n \mid n \in \mathbb{N}^+\}$ in $\mathbb{R}$. Consider $\epsilon > 0$. We want to find $a \in \{f_n \mid n \in \mathbb{N}^+\}$ so that $a > 2 - \epsilon$. To do so, we find $n \in \mathbb{N}^+$ so that $2 - \frac{1}{2^n} > 2 - \epsilon$.

Simplifying this expression, we seek $n \in \mathbb{N}$ so that

$$\frac{1}{2^n} < \epsilon$$

By the Archimedian Principle there exists $n \in \mathbb{N}^+$ so that $\frac{1}{n} < \epsilon$. Since $2^n > n$ (which we can prove by induction),

$$\frac{1}{2^n} < \frac{1}{n} < \epsilon$$

Therefore there exists $a \in \{f_n \mid n \in \mathbb{N}^+\}$ so that $a > 2 - \epsilon$.

(d) No, we have not (yet) defined an infinite sum as the limit of the sequence of partial sums.